MSG No. 135.486 $P4'_2/mb'c$ [Type III, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `2/m..`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,5,6]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,4,7,8]`
3	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[9,10,13,14]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[11,12,15,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `-4'..`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,13,14]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[3,4,15,16]`
3	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[5,6,9,10]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[7,8,11,12]`

Table 3: Wyckoff site: `4c`, site symmetry: `2/m..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,2,5,6]`
2	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,4,7,8]`
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[9,10,13,14]`
4	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[11,12,15,16]`

Table 4: Wyckoff site: `4d`, site symmetry: `2.22`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,2,3,4]`
2	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[5,6,7,8]`
3	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[9,10,11,12]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[13,14,15,16]`

Table 5: Wyckoff site: `8e`, site symmetry: `2..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[3,4]`
3	$[0,\ 0,\ -z]$	`[5,6]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[7,8]`
5	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[9,10]`
6	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[11,12]`
7	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[13,14]`

8	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[15,16]`

Table 6: Wyckoff site: `8f`, site symmetry: `2..`

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	`[1,2]`
2	$\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$	`[3,4]`
3	$\left[0, \frac{1}{2}, -z\right]$	`[5,6]`
4	$\left[0, \frac{1}{2}, z + \frac{1}{2}\right]$	`[7,8]`
5	$\left[\frac{1}{2}, 0, z + \frac{1}{2}\right]$	`[9,10]`
6	$\left[\frac{1}{2}, 0, -z\right]$	`[11,12]`
7	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	`[13,14]`
8	$\left[\frac{1}{2}, 0, z\right]$	`[15,16]`

Table 7: Wyckoff site: `8g`, site symmetry: `..2`

No.	position	mapping
1	$\left[x, x + \frac{1}{2}, \frac{1}{4}\right]$	`[1,3]`
2	$\left[-x, \frac{1}{2} - x, \frac{1}{4}\right]$	`[2,4]`
3	$\left[-x, \frac{1}{2} - x, \frac{3}{4}\right]$	`[5,7]`
4	$\left[x, x + \frac{1}{2}, \frac{3}{4}\right]$	`[6,8]`
5	$\left[\frac{1}{2} - x, x, \frac{3}{4}\right]$	`[9,12]`
6	$\left[x + \frac{1}{2}, -x, \frac{3}{4}\right]$	`[10,11]`
7	$\left[x + \frac{1}{2}, -x, \frac{1}{4}\right]$	`[13,16]`
8	$\left[\frac{1}{2} - x, x, \frac{1}{4}\right]$	`[14,15]`

Table 8: Wyckoff site: `8h`, site symmetry: `m..`

No.	position	mapping
1	$[x, y, 0]$	`[1,6]`
2	$[-x, -y, 0]$	`[2,5]`
3	$\left[y + \frac{1}{2}, x + \frac{1}{2}, \frac{1}{2}\right]$	`[3,8]`
4	$\left[\frac{1}{2} - y, \frac{1}{2} - x, \frac{1}{2}\right]$	`[4,7]`
5	$\left[-y, x, \frac{1}{2}\right]$	`[9,14]`
6	$\left[y, -x, \frac{1}{2}\right]$	`[10,13]`
7	$\left[x + \frac{1}{2}, \frac{1}{2} - y, 0\right]$	`[11,16]`
8	$\left[\frac{1}{2} - x, y + \frac{1}{2}, 0\right]$	`[12,15]`

Table 9: Wyckoff site: `16i`, site symmetry: `1`

No.	position	mapping
1	$[x, y, z]$	`[1]`
2	$[-x, -y, z]$	`[2]`
3	$\left[y + \frac{1}{2}, x + \frac{1}{2}, \frac{1}{2} - z\right]$	`[3]`

continued ...

Table 9

No.	position	mapping
4	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[4]
5	$[-x,\ -y,\ -z]$	[5]
6	$[x,\ y,\ -z]$	[6]
7	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[7]
8	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[8]
9	$\left[-y,\ x,\ z+\frac{1}{2}\right]$	[9]
10	$\left[y,\ -x,\ z+\frac{1}{2}\right]$	[10]
11	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[11]
12	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ -z\right]$	[12]
13	$\left[y,\ -x,\ \frac{1}{2}-z\right]$	[13]
14	$\left[-y,\ x,\ \frac{1}{2}-z\right]$	[14]
15	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	[15]
16	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	[16]